

Related Rates — Problem Set

Note: The problems may assume you have covered trigonometry and volumes of revolution

Warm-up

1. If $y = x^2 + 5x$ and $\frac{dx}{dt} = 3$, find $\frac{dy}{dt}$ when $x = 2$.
2. A square has side length s cm. If the side length is increasing at a rate of 2 cm s^{-1} , find the rate at which the area A is increasing when $s = 10$ cm.
3. The volume of a cube is given by $V = x^3$. Find an expression for $\frac{dV}{dt}$ in terms of x and $\frac{dx}{dt}$.
4. A circle has radius r . If the area A is increasing at $10\pi \text{ cm}^2\text{s}^{-1}$, find $\frac{dr}{dt}$ when $r = 5$ cm.
5. The volume of a sphere is $V = \frac{4}{3}\pi r^3$. Show that $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.
6. Given $V = \frac{1}{3}\pi r^2 h$ and h is constant, find $\frac{dV}{dt}$ when $r = 3$, $h = 4$ and $\frac{dr}{dt} = 0.5$.
7. A rectangle has a fixed width of 10 cm. If its length l is increasing at 3 cm s^{-1} , find the rate of change of its perimeter P .
8. If $x^2 + y^2 = 25$ and $\frac{dx}{dt} = 4$, find $\frac{dy}{dt}$ at the point $(3, 4)$.
9. The surface area of a cube is $S = 6x^2$. If x is increasing at 0.1 cm s^{-1} , find $\frac{dS}{dt}$ when $x = 5$ cm.
10. Water is leaking from a tank such that the volume V decreases at a rate of 2 L/min . State the value of $\frac{dV}{dt}$ including the correct sign.

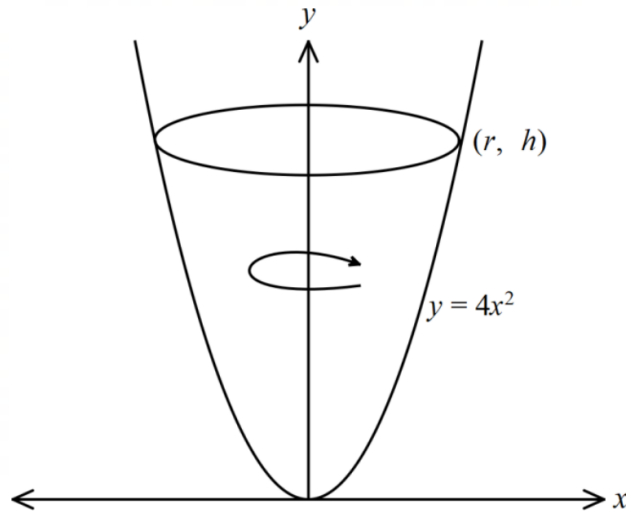
Skill-building

1. A spherical balloon is being inflated at a constant rate of $100 \text{ cm}^3\text{s}^{-1}$. Find the rate at which the surface area is increasing when the radius is 10 cm.
2. A ladder 5 m long leans against a vertical wall. If the bottom of the ladder slides away from the wall at 0.5 ms^{-1} , find the speed at which the top of the ladder is sliding down the wall when the bottom is 3 m from the wall.
3. A stone is dropped into a still pond, sending out circular ripples. The radius of the outer ripple increases at 2 cm s^{-1} . Find the rate at which the area of the disturbed water is increasing after 5 seconds.
4. The volume of a right circular cone is $V = \frac{1}{3}\pi r^2 h$. If the height is always twice the radius, find the rate of change of the volume with respect to time when the radius is 3 cm and is increasing at 1.5 cm s^{-1} .
5. A cylindrical tank with radius 2 m is being filled with water at a rate of $3 \text{ m}^3\text{min}^{-1}$. Find the rate at which the depth of the water is increasing.

6. A man 1.8 m tall walks away from a lamp post 6 m high at a speed of 1.2 ms^{-1} . Find the rate at which the length of his shadow is increasing.
7. Two cars start from the same point. Car A travels North at 60 km/h and Car B travels East at 80 km/h. Find the rate at which the distance between them is increasing after 2 hours.
8. The surface area of a melting ice cube decreases at a constant rate of $12 \text{ mm}^2\text{s}^{-1}$. Find the rate at which the volume is decreasing when the side length is 10 mm.
9. A trough is 10 m long and its ends are isosceles triangles with height 2 m and width 4 m at the top. If water is being pumped in at $0.5 \text{ m}^3\text{min}^{-1}$, find the rate at which the water level is rising when the water is 1 m deep.

Easier Exam Questions

1. (Glenwood 2021 Q13d) Water is poured into a container at a rate of $8 \text{ cm}^3\text{s}^{-1}$ and the volume of the water in the container is given by $V = \frac{3}{2}(h^2 + 8h)$ where h cm is the depth of the water. Find the rate of change of the depth of the water when $V = 72 \text{ cm}^3$. **3**
2. (Abbotsleigh 2025 Q12b) The volume of a cylinder with a fixed height h cm, and radius r cm is given by $V = \pi r^2 h$. If the volume is increasing at a rate of $20 \text{ cm}^3 \text{ s}^{-1}$, show that $\frac{dr}{dt} = \frac{10}{\pi r h} \text{ cm s}^{-1}$. **2**
3. (St George Girls 2022 Q13b) The volume of a cube is expanding at the constant rate of $5 \frac{\text{mm}}{\text{s}}$. At what rate is the surface area of the cube increasing when the side length of the cube is 60cm? **3**
4. (Fort St 2023 Q4a) A spherical ball is expanding so that its volume is increasing at the constant rate of 10 mm^3 per second. What is the rate of increase of the radius when the surface area is 400 mm^2 ? **2**
5. (Caringbah 2024 Q14c) The interior surface of a bowl is made by rotating the curve $y = \frac{x^2}{4}$ about the y -axis, the units on both axes being in centimetres (cm). Water is poured into the bowl at a constant rate of 36 cm^3 per second. When the depth of the water is h cm, the area of the surface of the water is $S \text{ cm}^2$. Find:
 - (i) the rate of increase in h when $S = 12$. **2**
 - (ii) the rate of increase in S when $h = 9$. **2**
6. (Pymble 2025 Q12e)
 - (i) The internal surface of a water reservoir is created by rotating the parabola about the y axis between $y = 0$ and $y = h$. **2**
Show that the volume of the water in the reservoir is given by $V = 2\pi r^4$.
 - (ii) The variable h represents the depth of the water in the reservoir at time t hours and r metres is the radius of the upper surface of the water at that time. **1**
The radius of the surface is reducing at a constant rate such that $\frac{dr}{dt} = -2$.
Show that the rate of change of the depth is given by $\frac{dh}{dt} = -16r$.
 - (iii) Find the rate of change of the volume when the radius is 5 metres. **2**

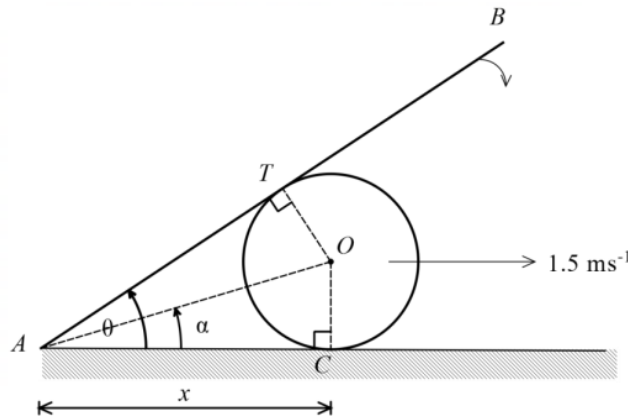


Harder Exam Questions

1. (Trinity Grammar 2024 Q14b)

A rod AB is leaning on a metal disc of radius 1 m, as shown in the diagram below. The centre of the disc is at O .

One end of the rod, A , is fixed to the ground, while the disc is tangential to the ground at C , such that $\angle ACO = \angle ATO = 90^\circ$.



The disc rolls horizontally to the right with a constant speed of 1.5 ms^{-1} . As the disc rolls the end of the rod in the air, B , descends to the ground.

Let $\angle OAC = \alpha$, $\angle BAC = \theta$ and $AC = x$.

(i) Write an expression for $\tan \alpha$ in terms of x .

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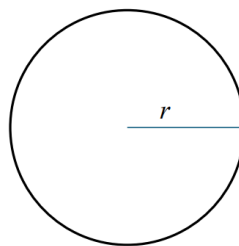
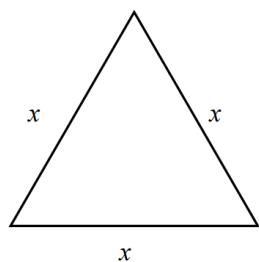
(ii) Hence, or otherwise, find the rate of change of θ in radians per second when $x = 4 \text{ m}$, giving your answer correct 3 decimal places.

4

2. (Glenwood 2023 Q14b)

PQ and SR are two towers of height 0.7 km. Q is 10 km due North of R . A vehicle at T is travelling due East away from Q at a constant speed of 10 km/h. Let the distance QT be x , the distance PT be k and $\angle PTS = \theta$.

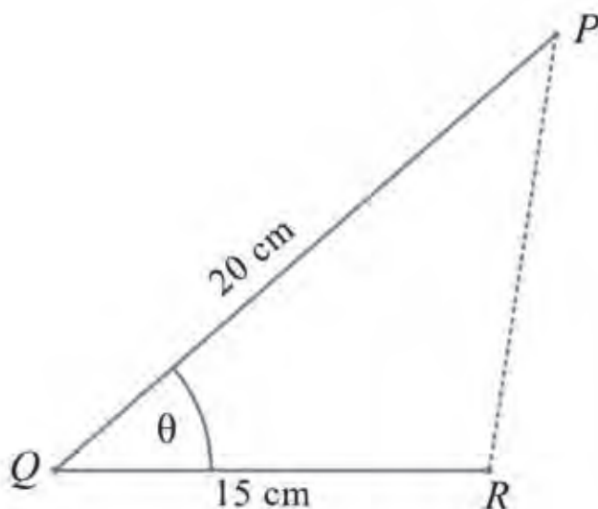
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5. (Merewether 2025 Q14b) The triangle PQR , shown below, has two sides of fixed length: $PQ = 20$ cm and $QR = 15$ cm. $\angle PQR = \theta$ is increasing at a rate of $\frac{\pi}{6}$ radians/hour.

Find the rate at which the area of the triangle is changing at the time when $\theta = \frac{\pi}{3}$.

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6. (Penrith 2022 Q14b)

The chord AC cuts off a minor segment ABC in a circle with centre O and radius y cm. $\angle AOC$ is x radians.

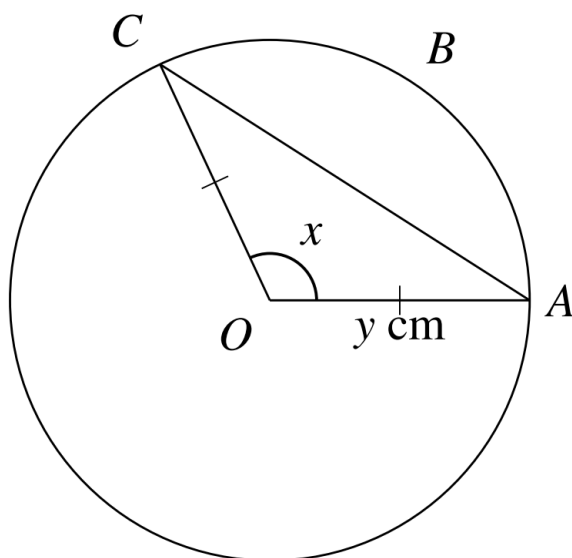
It is given that y and x vary so that the area of the minor segment is constant at 50 cm².

(i) Show that $y = \frac{10}{\sqrt{x - \sin x}}$

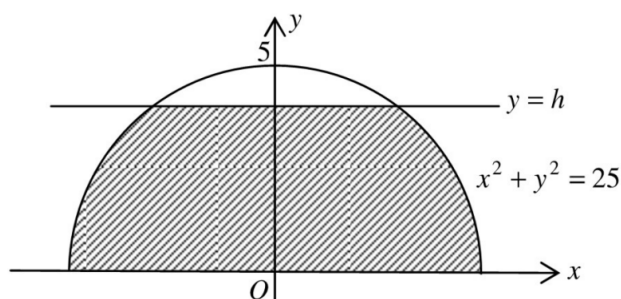
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- (ii) If x is increasing at a rate of 0.05 radians per second, find the rate at which the radius is decreasing when $x = 1.5$ radians, correct to 2 decimal places.

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7. (Sydney Boys 2024 Q14b)

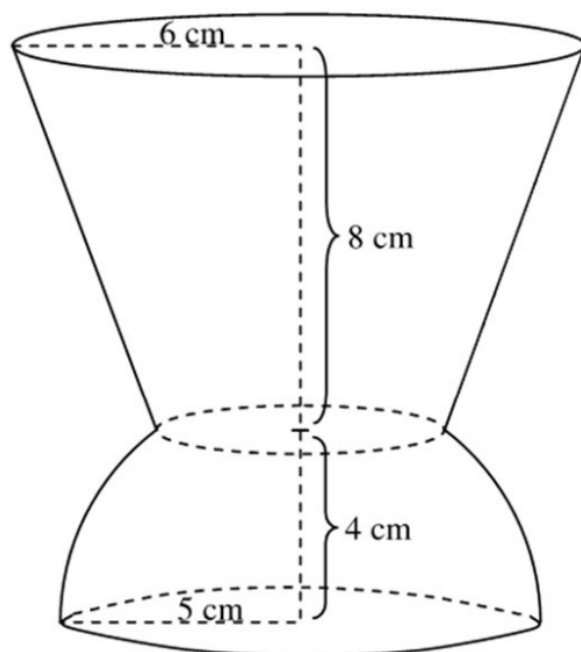


(i) In the figure above, the shaded region enclosed by the circle $x^2 + y^2 = 25$, the x -axis, and the straight line $y = h$ (where $0 \leq h \leq 5$) is revolved about the y -axis.

Show that the volume of the solid of revolution is $\left(25h - \frac{h^3}{3}\right) \pi$ cubic units.

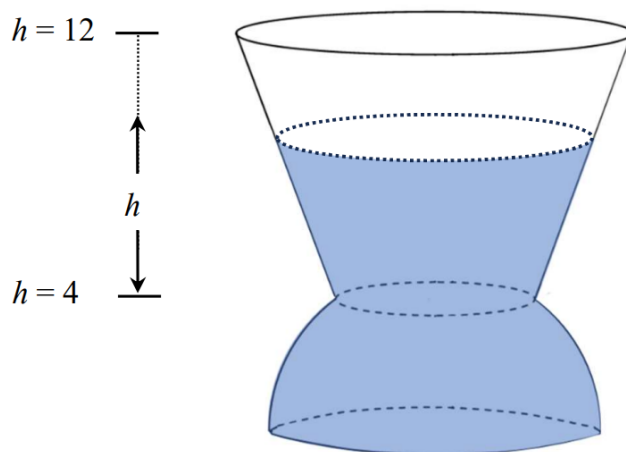
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In the diagram below, an empty coffee cup consists of two parts. The lower part is in the shape of the solid described in (b) (i). The upper part is a frustum of a circular cone of height 8 cm, where the radius of the top of the cup is 6 cm. Hot coffee is poured into the cup to a depth h cm at a rate of $8 \text{ cm}^3/\text{s}$, where $0 \leq h \leq 12$. Let $V \text{ cm}^3$ be the volume of coffee in the cup.



- (ii) Find the rate of increase of the depth of coffee when the depth is 3 cm. 2
- (iii) Show that the volume of coffee in the cup when for $4 \leq h \leq 12$ is given by 3

$$V = \frac{164\pi}{3} + \frac{3\pi}{64}(h+4)^3.$$



- (iv) After the cup is fully filled, it cracks at the bottom. The coffee leaks at a rate of $2 \text{ cm}^3/\text{s}$. Find the rate of decrease of the depth of coffee after 15 seconds of leaking. Leave your answer correct to 3 significant figures. 3